



Addis Ababa University
Department of Computer Science
AgriNet
Test Plan Document

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1. Introduction

AgriNet is a comprehensive digital agricultural marketplace platform developed by Addis Ababa University students. The system connects farmers directly with buyers, provides an expert advisory portal, supports real-time chat between users, and includes an admin dashboard for platform governance. This Test Plan Document describes the testing strategy, test cases, and acceptance criteria used to validate the AgriNet system against its documented functional and non-functional requirements.

1.1 Purpose

The purpose of this test plan is to:

- ❖ Uncover and report as many defects as possible to increase system reliability.
- ❖ Ensure the right product is developed in accordance with the specified functional and non-functional requirements.
- ❖ Verify that all user roles Farmer, Buyer, Agricultural Expert, and Admin can perform their designated functions correctly and securely.
- ❖ Validate that the system meets its non-functional requirements including usability, performance, security, and compatibility across devices and browsers.
- ❖ Provide a structured, traceable test record for the AgriNet platform that can be used for future regression testing and maintenance.

2. Features to be Tested / Not to be Tested

2.1. Features to be Tested

This section lists the features to be tested from the user's viewpoint. Risk levels are rated as H (High), M (Medium), and L (Low).

User Management (FR1, FR2)

- ❖ **User Registration:** Create new accounts for Farmer, Buyer, and Expert roles — Risk: H
- ❖ **User Login / Authentication:** Secure login with email and password — Risk: H
- ❖ **Manage User Profile:** Edit personal details and account settings — Risk: M
- ❖ **Role-based Authorization:** Ensure each role can only access permitted features — Risk: H

Product Listing & Marketplace (FR3, FR4)

- ❖ **Post Product:** Farmer adds a new product for sale — Risk: H
- ❖ **Edit Product:** Farmer updates an existing product listing — Risk: M
- ❖ **Delete Product:** Farmer removes a product from the marketplace — Risk: M
- ❖ **Search & Browse Products:** Buyer searches and filters the product catalog — Risk: H
- ❖ **View Product Details:** Buyer views full details of a listed product — Risk: M

Cart & Order Management (FR5)

- ❖ **Add to Cart:** Buyer adds products to the shopping cart — Risk: H
- ❖ **Place Order:** Buyer completes a purchase from the cart — Risk: H
- ❖ **View Orders:** Farmer and Buyer view order history and status — Risk: M

Expert Advisory Portal (FR6)

- ❖ **Expert Publish Article/Guide:** Expert posts agricultural advisory content — Risk: M
- ❖ **Expert Respond to Farmer Query:** Expert answers submitted questions — Risk: M
- ❖ **Access Expert Advisory:** Farmer reads guides and submits queries — Risk: M

Communication (FR3–FR6)

- ❖ **Chat Between Buyer and Seller:** Real-time messaging between users — Risk: M
- ❖ **Report a User:** Any user reports inappropriate behaviour to admin — Risk: M

Admin Dashboard (FR7)

- ❖ **Admin Manage Users:** Admin verifies, suspends, or removes user accounts — Risk: H
- ❖ **Admin Monitor System Analytics:** Admin views dashboard and reports — Risk: M

Non-Functional Requirements to be Tested

- ❖ **User Interface:** Intuitive navigation, mobile-first responsive design, accessible visual design.
- ❖ **Hardware and Software:** Cross-browser (Chrome, Firefox, Edge, Safari) and device (mobile, tablet, desktop) compatibility.
- ❖ **Security:** Authentication, authorization, input validation, data encryption.
- ❖ **Performance:** Page load times under 500ms threshold; responsiveness under concurrent users.

2.2. Features Not to be Tested

The following features are not in scope for this test plan:

- ❖ **Transportation/Logistics Module:** Relies on third-party providers not fully integrated in this release.
- ❖ **Live Market Price Feed Integration:** Full integration with national agricultural market systems is not implemented.
- ❖ **Payment Gateway Processing:** Actual financial transaction processing through external gateways is beyond project scope.

- ❖ Demand Forecasting and Advanced Analytics: Identified as future features, not implemented in this release.
- ❖ Video Tutorial Playback: Multimedia content delivery is not implemented in this release.

3. Pass/Fail Criteria

The success/fail criteria are the following:

- ❖ PASS: The actual output produced by the system matches the expected result defined in the test case specification.
- ❖ FAIL: The actual output differs from the expected result, including incorrect messages, missing validations, system errors, crashes, or security violations.

Additional criteria:

- ❖ A test case that causes an unhandled system exception or crash is automatically marked FAIL.
- ❖ A test case involving security (e.g. unauthorized access, SQL injection) is FAIL if the system does not properly block the attempt.
- ❖ All HIGH-risk features must achieve 100% PASS before release. MEDIUM and LOW risk features must achieve at least 90% PASS rate.

4. Approach/Strategy

This section describes the overall approach and strategy for testing the AgriNet system. The project follows the Waterfall SDLC methodology; testing is conducted after development using a structured, requirement-driven approach.

Testing Levels:

- ❖ Unit Testing: Individual functions and components tested in isolation by developers using Jest.
- ❖ Integration Testing: Interactions between subsystems tested (e.g. Cart with Order module, Marketplace with PostgreSQL via Prisma ORM).
- ❖ System Testing: The fully integrated AgriNet system tested end-to-end against all functional requirements.
- ❖ User Acceptance Testing (UAT): Representative end users (farmers, buyers) validate the system meets real-world needs.

Testing Types:

- ❖ Functional Testing: Verifying all functional requirements (FR1-FR7) as listed in Section 2.1.
- ❖ Usability Testing: Assessing the interface against NFRs navigation, mobile responsiveness, accessibility.
- ❖ Security Testing: Authentication, authorization, input validation (SQL injection, XSS), data encryption.
- ❖ Performance Testing: Measuring response times to verify the 500ms interaction threshold.

- ❖ Compatibility Testing: Verifying correct rendering across Chrome, Firefox, Edge, and Safari on mobile and desktop.
- ❖ Regression Testing: Re-running affected test cases after any bug fix or change.

Tools:

- ❖ Jest: Unit and integration testing for JavaScript/Next.js components.
- ❖ Postman: API endpoint testing for backend services.
- ❖ Browser DevTools: Manual compatibility and performance checks.
- ❖ pgAdmin / Prisma Studio: Database integrity and data validation.
- ❖ GitHub: Version control — test scripts committed alongside source code.

Configuration Management:

- ❖ All test cases are version-controlled in the project repository.
- ❖ Two configurations tested: (1) Development local Next.js server with local PostgreSQL; (2) Staging cloud-hosted deployment mirroring production.
- ❖ Hardware: Standard laptops (Windows/macOS) and Android/iOS smartphones for mobile testing.
- ❖ Software: Node.js, Next.js, PostgreSQL, Prisma ORM, modern browsers.

Metrics to be Collected:

- ❖ Total test cases executed vs. planned.
 - ❖ Number of PASS / FAIL per feature and per risk level.
 - ❖ Defect density: number of defects found per feature module.
 - ❖ Test coverage: percentage of functional requirements covered by test cases.
5. Test Cases with Specifications

A test case is a set of input data and expected results that exercises a component with the purpose of causing failures and detecting faults. The following test cases cover all functional requirements identified in Section 2.1.

Table 1: Test Case Specification for User Registration

Name: User Registration				
Purpose: To allow Farmers, Buyers, and Experts to create a verified account on AgriNet.				
Test Data: Full Name Email Password Confirm Password Role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Empty Full Name; all other fields valid	"Full name is required"	Empty name, valid email/password/role	"Full name is required"	Fail
Empty Email; all other fields valid	"Email is required"	Empty email, valid other fields	"Email is required"	Fail
Invalid Email format	"Enter a valid email address"	Email="user@", valid other fields	"Enter a valid email"	Fail
Already registered email	"Email already in use"	Existing email, valid other fields	"Email already in use"	Fail
Password less than 8 characters	"Password must be at least 8 characters"	Password=abc123	"Password too short"	Fail
Password / Confirm Password mismatch	"Passwords do not match"	Password=Pass1234, Confirm=Pass5678	"Passwords do not match"	Fail
Empty Password; all other fields valid	"Password is required"	Empty password, valid other fields	"Password is required"	Fail
No role selected; all other fields valid	"Please select a role"	Valid fields; no role chosen	"Please select a role"	Fail
All fields empty	"All fields are required"	All inputs empty	"All fields are required"	Fail
All valid fields — Farmer role	"Registration successful. Welcome to AgriNet!"	Valid name, email, strong password, role=Farmer	"Registration successful"	Pass

Table 2: Test Case Specification for User Login

Name: User Login				
Purpose: To authenticate registered users and grant access to role-specific features.				
Test Data: Email (registered/unregistered/empty) Password (correct/incorrect/empty)				
Input	Expected Result	Data	Actual Output	Pass/Fail
Empty email; any password	"Email is required"	Empty email field	"Email is required"	Fail
Empty password; any email	"Password is required"	Valid email, empty password	"Password is required"	Fail
Both fields empty	"All fields are required"	Both inputs empty	"All fields are required"	Fail
Unregistered email; any password	"No account found with this email"	Email not in system	"No account found"	Fail
Registered email; incorrect password	"Incorrect password"	Correct email, wrong password	"Incorrect password"	Fail
Valid credentials — Farmer	Redirected to Farmer Dashboard	Valid Farmer credentials	Farmer Dashboard loaded	Pass
Valid credentials — Buyer	Redirected to Buyer Dashboard	Valid Buyer credentials	Buyer Dashboard loaded	Pass
Valid credentials — Expert	Redirected to Expert Dashboard	Valid Expert credentials	Expert Dashboard loaded	Pass
Valid credentials — Admin	Redirected to Admin Dashboard	Valid Admin credentials	Admin Dashboard loaded	Pass
SQL injection in email field	Input rejected; no system error	Email = ' OR '1'=1	Input rejected; login blocked	Fail

Table 3: Test Case Specification for Manage User Profile

Name: Manage User Profile				
Purpose: To allow any registered user to view and update their profile information.				
Test Data: Full Name Email Profile Photo (valid image / invalid type / oversized)				
Input	Expected Result	Data	Actual Output	Pass/Fail
Empty Full Name; all other fields valid	"Full name is required"	Clear name field, submit	"Full name is required"	Fail
Invalid email format	"Enter a valid email address"	Email="notanemail"	"Enter a valid email"	Fail
Email already used by another account	"Email is already in use by another account"	Existing email of another user	"Email already in use"	Fail
Upload image exceeding size limit	"Image must be under 2MB"	Upload a 5MB image	"Image must be under 2MB"	Fail
All valid fields; submit update	"Profile updated successfully"	Valid name, email, image	"Profile updated successfully"	Pass

Table 4: Test Case Specification for Post Product (Farmer)

Name: Post Product (Farmer)				
Purpose: To allow a verified Farmer to list a new agricultural product for sale on the marketplace.				
Test Data: Product Name Category Price Quantity Image				
Input	Expected Result	Data	Actual Output	Pass/Fail
Empty product name; all other fields valid	"Product name is required"	Empty name, valid other fields	"Product name is required"	Fail
No category selected	"Please select a category"	Valid name/price/qty, no category	"Please select a category"	Fail
Quantity = 0	"Quantity must be at least 1"	Qty=0, valid other fields	"Quantity must be at least 1"	Pass
All required fields empty	"All required fields must be filled"	All inputs empty	"All required fields must be filled"	Fail
Valid fields; no image	Product listed with default image	Valid name, category, price, qty; no image	Product listed successfully	Pass
Valid fields; with image	"Product listed successfully"	Valid all fields + valid image	"Product listed successfully"	Pass
Non-farmer user attempts to post	Access denied / redirected to login	Buyer/Expert tries to access /farmer/post	Access denied	Fail

Table 5: Test Case Specification for Edit Product

Name: Edit Product (Farmer)				
Purpose: To allow a Farmer to update the details of an existing product listing they own.				
Test Data: Product Name Price Quantity Unauthorized access attempts				
Input	Expected Result	Data	Actual Output	Pass/Fail
Farmer edits own product with valid data	"Product updated successfully"	Valid updated fields for own listing	"Product updated successfully"	Pass
Farmer clears the product name field	"Product name is required"	Delete name and submit	"Product name is required"	Pass
Farmer attempts to edit product not owned by them	Access denied	Edit URL of another farmer's product	Access denied	Fail
Buyer attempts to access edit product page	Access denied / redirected	Buyer session accesses edit route	Access denied / redirected	Pass

Table 6: Test Case Specification for Delete Product

Name: Delete Product (Farmer)				
Purpose: To allow a Farmer to permanently remove one of their product listings from the marketplace.				
Test Data: Product owned by Farmer Product with active orders Unauthorized user				
Input	Expected Result	Data	Actual Output	Pass/Fail
Farmer deletes own product; no active orders	Confirmation prompt shown; product removed on confirm	Valid delete request on own product	Product removed from marketplace	Pass
Farmer cancels delete on confirmation dialog	Product remains listed; no changes	Click Delete then Cancel	Product remains listed	Pass
Farmer deletes product with pending orders	"Cannot delete a product with active orders"	Delete product with undelivered orders	"Cannot delete — active orders exist"	Fail
Farmer attempts to delete another farmer's product	Access denied	Direct DELETE API call with wrong farmer ID	Access denied	Fail
Admin deletes any product from dashboard	Product removed; system log updated	Admin delete action on any listing	Product removed; log entry created	Pass

Table 7: Test Case Specification for Search and Browse Products (Buyer)

Name: Search & Browse Products (Buyer)				
Purpose: To allow a Buyer to find agricultural products using keyword search and category/price filters.				
Test Data: Search Query (matching/non-matching/empty) Category Filter Price Range Filter				
Input	Expected Result	Data	Actual Output	Pass/Fail
Valid keyword matching existing products	List of matching products displayed	Search="Tomato"	Matching products shown	Pass
Valid keyword with no matching products	"No products found for your search"	Search="XYZ999"	"No products found"	Pass
Empty search with no filters applied	All available products displayed (paginated)	Submit empty search	Full product catalog shown	Pass
Filter by valid category	Only products in selected category shown	Category=Vegetables	Filtered product list displayed	Pass
Filter by valid price range	Products within the price range shown	Min=50, Max=200	Filtered product list displayed	Pass
Invalid price range (Min > Max)	"Minimum price cannot exceed maximum price"	Min=500, Max=100	"Min cannot exceed Max"	Fail
Combine keyword + category + price filter	Products matching all criteria shown	Search=Maize, Category=Grains, Price 100-500	Correctly filtered list shown	Pass
Unauthenticated user browses marketplace	Products displayed; purchase requires login	Guest accesses marketplace	Products visible; login prompt on purchase	Pass

Table 8: Test Case Specification for Add to Cart

Name: Add to Cart (Buyer)				
Purpose: To allow an authenticated Buyer to add one or more products to their shopping cart.				
Test Data: Product availability (in stock/out of stock) Quantity (valid/zero/exceeds stock/text) User role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Add in-stock product with valid quantity	Product added to cart; cart count updated	Qty=2, in-stock product	Cart updated with item and quantity	Pass
Add product with quantity exceeding stock	"Only X items available in stock"	Stock=5, Qty requested=10	"Only 5 items available in stock"	Fail
Add product with non-numeric quantity	"Enter a valid quantity"	Qty="abc"	"Enter a valid quantity"	Fail
Add out-of-stock product	"This product is currently out of stock"	Click Add to Cart on out-of-stock item	"Product is currently out of stock"	Fail
Unauthenticated user clicks Add to Cart	Redirected to login page	Guest user clicks Add to Cart	Login page shown	Fail
Farmer attempts to add a product to cart	Access restricted; cart is for Buyers only	Farmer session tries to add to cart	Access restricted message shown	Fail
Add same product to cart a second time	Quantity incremented in existing cart item	Add product already in cart	Quantity updated in cart	Fail

Table 9: Test Case Specification for Place Order

Name: Place Order (Buyer)				
Purpose: To allow an authenticated Buyer to confirm and place an order for items in their cart.				
Test Data: Cart state (items present/empty) Delivery address (valid/empty) Product availability at checkout				
Input	Expected Result	Data	Actual Output	Pass/Fail
Place order with valid cart items and address	"Order placed successfully! Order ID: #XXXX"	Cart has items, valid delivery address	"Order placed successfully"	Pass
Attempt to place order with empty cart	"Your cart is empty"	Cart is empty, click Place Order	"Your cart is empty"	Pass
Place order without entering delivery address	"Delivery address is required"	Valid cart, empty address field	"Delivery address is required"	Pass
Product became out of stock at checkout	"One or more items in your cart are no longer available"	Product sold out between add-to-cart and checkout	"Item no longer available"	Pass
Unauthenticated user accesses checkout	Redirected to login	Guest user navigates to /checkout	Login page shown	Fail
Buyer views order history after placing order	Order appears in history with correct details	Navigate to My Orders after successful order	Order listed with ID, date, status	Pass

Table 10: Test Case Specification for Chat Between Buyer and Seller

Name: Chat Between Buyer and Seller				
Purpose: To allow real-time messaging between a Buyer and a Farmer (seller) regarding a product.				
Test Data: Message content (valid/empty/very long) Sender role (Buyer/Farmer/unauthenticated)				
Input	Expected Result	Data	Actual Output	Pass/Fail
Buyer sends valid text message to Farmer	Message delivered; appears in both chat windows	Valid message sent by authenticated Buyer	Message visible to both parties	Pass
Buyer sends empty message	"Message cannot be empty"	Empty input, click Send	"Message cannot be empty"	Pass
Buyer sends message over 1000 characters	"Message is too long (max 1000 characters)"	Type 1001-char message, send	"Message is too long"	Fail
Farmer replies to Buyer's message	Reply appears in chat thread in correct order	Farmer sends reply from their side	Reply shown correctly in conversation	Pass
Unauthenticated user attempts to access chat	Redirected to login	Guest accesses /chat route	Login page shown	Fail

Table 11: Test Case Specification for Access Expert Advisory

Name: Access Expert Advisory (Farmer)				
Purpose: To allow a Farmer to browse expert articles, guides, and market alerts, and to submit queries to experts.				
Test Data: Query text (valid/empty) Article availability User role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Farmer browses available articles/guides	List of published articles/guides displayed	Navigate to Expert Advisory section	Articles and guides shown	Pass

Farmer views a specific article	Full article content displayed with author and date	Click on a published article	Article content rendered correctly	Pass
No articles published yet	"No advisory content available at this time"	Expert advisory section with no posts	"No content available"	Fail
Farmer submits valid query to expert	"Your query has been submitted successfully"	Type valid question, submit	"Query submitted successfully"	Pass
Farmer submits empty query	"Query cannot be empty"	Empty query field, click Submit	"Query cannot be empty"	Fail
Unauthenticated user accesses expert advisory	Public articles visible; query requires login	Guest navigates to expert advisory	Articles shown; login prompt on submit	Fail

Table 12: Test Case Specification for Report a User

Name: Report a User				
Purpose: To allow any authenticated user to report another user for inappropriate behaviour for admin review.				
Test Data: Report reason (selected/not selected) Description (optional) Reporter role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Submit report with a valid reason selected	"Report submitted. Our team will review it."	Select a reason, click Submit	"Report submitted successfully"	Pass
Submit report without selecting a reason	"Please select a reason for the report"	No reason selected, click Submit	"Please select a reason"	Fail
Submit report with optional description	Report logged with description attached	Select reason + type description, submit	Report stored with all details	Pass
Unauthenticated user attempts to report	Redirected to login	Guest accesses report feature	Login page shown	Fail

User reports themselves	"You cannot report your own account"	User selects own profile and reports	"Cannot report your own account"	Fail
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Table 13: Test Case Specification for Admin Manage Users

Name: Admin Manage Users				
Purpose: To allow the Admin to view, verify, suspend, or remove any user account from the platform.				
Test Data: User status (active/suspended/unverified) Action (verify/suspend/delete) Admin vs. non-admin role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Admin verifies an unverified seller	User status updated to Verified; user notified	Admin clicks Verify on pending seller	Seller status = Verified	Pass
Admin suspends an active user	User account suspended; user cannot log in	Admin clicks Suspend on active user	Account suspended	Pass
Admin deletes a user account	Account removed after confirmation	Admin clicks Delete; confirms dialog	Account removed from system	Pass
Admin cancels delete on confirmation dialog	No changes made	Click Delete then Cancel	User account unchanged	Pass
Admin searches for a user by name/email	Matching user(s) displayed in results	Enter name/email in admin search	Correct user record shown	Pass
Non-admin user attempts to access admin dashboard	Access denied; redirected	Farmer/Buyer navigates to /admin route	Access denied	Fail
Admin views system analytics dashboard	Dashboard metrics displayed correctly	Admin navigates to Analytics section	Charts and stats rendered	Pass

Table 14: Test Case Specification for Expert Publish Article/Guide

Name: Expert Publish Article/Guide				
Purpose: To allow an Agricultural Expert to publish informational articles, guides, or market alerts for farmers.				
Test Data: Title (valid/empty) Content (valid/empty) Category (selected/not selected) Author role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Expert publishes article with all valid fields	"Article published successfully"	Valid title, content, category selected	"Article published successfully"	Pass
Expert submits with empty title	"Title is required"	Empty title field, valid content	"Title is required"	Fail
Expert submits with empty content	"Article content cannot be empty"	Valid title, empty content body	"Content cannot be empty"	Fail
Expert submits with no category selected	"Please select a category"	Valid title and content, no category	"Please select a category"	Fail
Farmer attempts to publish an article	Access denied; Expert-only feature	Farmer session accesses publish route	Access denied	Fail
Expert updates a previously published article	"Article updated successfully"	Edit existing article with valid changes	"Article updated successfully"	Pass
Expert deletes one of their articles	"Article deleted successfully"	Click Delete on own article, confirm	"Article deleted successfully"	Pass

Table 15: Test Case Specification for Expert Respond to Farmer Query

Name: Expert Respond to Farmer Query				
Purpose: To allow an Agricultural Expert to view and respond to questions submitted by Farmers.				
Test Data: Response text (valid/empty) Query status (open/already answered) Responder role				
Input	Expected Result	Data	Actual Output	Pass/Fail
Expert submits valid response to open query	Response posted; Farmer notified	Valid response text, open query	Response visible to Farmer	Pass
Expert submits empty response	"Response cannot be empty"	Empty text, click Respond	"Response cannot be empty"	Fail
Expert responds to an already-answered query	Response added as additional clarification	Respond to query that already has one answer	Additional response shown	Pass
Farmer attempts to respond to any query	Access restricted; response is Expert-only	Farmer tries to post response to any query	Access restricted	Fail
Farmer views expert response to their own query	Response displayed under their submitted query	Farmer navigates to My Queries	Expert response shown correctly	Pass

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